Determinants of Insurance Premiums: Lifestyle Choices vs. Demographic Factors

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#AI Statement: We used AI to help suggest potential ways to compare regression models. Further, we used AI to provide us information on how we can conduct the test as well as how to format the code nicely. We also asked for potential suggestions on how we can interpret the data.

1. Introduction

1.1 Context and Research Question

In 2023, healthcare spending in the United States increased by 7.5%, totaling 4.9 trillion USD [1]. This increase in healthcare spending paralells increases in health insurance costs. As of recent health insurance costs have exceeded the growth rates of wages and inflation, making it increasingly difficult for individuals to afford coverage [2].

However, health insurance vary from individual to individual; according to the Affordable Care Act, factors such as age, location and tobacco use are legally allowed to increase the cost of one's premium [3]. Further, individuals with other pre-existing health risks may also select costlier healthcare plans. Overall, we can categorize factors that potentially drive healthcare cost into two categories: Modifiable Lifestyle Choices (MLC) (behaviors an individual can change) and Baseline Demographic/Socioeconomic Factors (DRF) (fixed characteristics).

Many prior studies have focused on investigating singular variables while others have focused on participation in healthcare as opposed to cost [8] [9]. This study will attempt to fill a gap by comparing the extent of variability in health insurance cost attributable to MLC and DRF: Does one category have a more substantial effect on insurance cost variance? The results may indicate how institutions and individuals can make healthcare more affordable. If insurance cost variance is heavily driven by MLC, customers can lower prices by modifying behaviors. However, if insurance cost is heavily driven by DRF, policy changes may be required to lower costs.

Research Question: **To what extent can variation in insurance premiums be attributed to Baseline Demographic Factors (DRF) versus Modifiable Lifestyle Factors (MLC)**

Dataset: We use the Medical Insurance Cost Prediction dataset from Kaggle [4]. The dataset contains 100,000 observations including demographics (Age, Sex, Income) and lifestyle indicators (Smoking status, BMI, Alcohol frequency). Data was derived from three main sources: Centers for Disease Control & Prevention (CDC), Department of Health & Human Services (HHS), and

Centers for Medicare & Medicaid Services (CMS). The response variable is the annual healthcare premium cost.

Modifiable Lifestyle Choices (MLC): Smoking status (Current, Former, Never), Alcohol consumption frequency, and Body Mass Index (BMI).

Baseline Demographics (DRF): Age, Sex, Education level, Income, Marital Status, and Household Size.

1.2 Exploratory Data Analysis

Table 1 provides summary statistics of the response variable (Annual Premium), namely, median, mean, minimum and maximum. As the mean premium, at 582.32 USD, is substantially higher than the median, at 463.58 USD, we can infer a right-skewed distribution. Further, the high value of the maximum premium, at 10962.55 USD, especially in comparison to the minimum premium value of 211.67 USD, may imply the presence of high outliers.

Table 1: Summary Statistics of Annual Premium Costs

Median Premium	Mean Premium	Min Premium	Max Premium
463.58	582.32	211.67	10962.55

2. Methodology

To compare the variance explained by lifestyle choices versus demographics, we employ Hierarchical Multiple Linear Regression [5]. This approach allows us to enter variables in blocks to determine the unique variance explained by each category. We utilize Analysis of Variance (ANOVA) using Partial F-tests to determine if the addition of a block of variables to a reduced model significantly reduces the Sum of Squared Errors (SSE) and thus R^2.

We fit three distinct models:

Model 1: Baseline Demographics Accounts only for fixed factors the individual cannot easily change.

$$\ln(Y) = \beta_0 + \beta_{Age}x_{Age} + \beta_{Sex}x_{Sex} + \beta_{Region}x_{Region} + \beta_{Income}x_{Income} + \cdots + \epsilon$$

Model 2: Lifestyle Only Accounts only for modifiable behaviors.

$$\ln(Y) = \ln(Y) = \beta_0 + \beta_{Smoker}x_{Smoker} + \beta_{Alcohol}x_{Alcohol} + \beta_{BMI}x_{BMI} + \epsilon$$

Model 3: Full Model (Demographics + Lifestyle) Combines both categories and includes the interaction term observed in EDA.

$$\ln(Y) = \text{Model 1} + \text{Model 2} + \epsilon$$

While ANOVA tests for statistical significance, it does not quantify effect size. Therefore, we also compare the change in Adjusted R^2 between the partial models (DRF vs. Full and MLC vs. Full). This allows us to assess which set of variables has a more substantial practical effect on reducing the squared residuals.

2.1 Log Transformation

The use of a log transformation is to address the requirement of normality in the residuals required by Least Squares Regression [6]. When a full model is fit without log, its associated residual histogram is right skewed (Figure 1, left). A log-transformation was applied [6] to the outcome variable: annual premium, resulting in a more symmetric residual distribution (Figure 1, right). Other models (DRF and MLC) also showed roughly normal distributions when log was applied to the outcome variable.

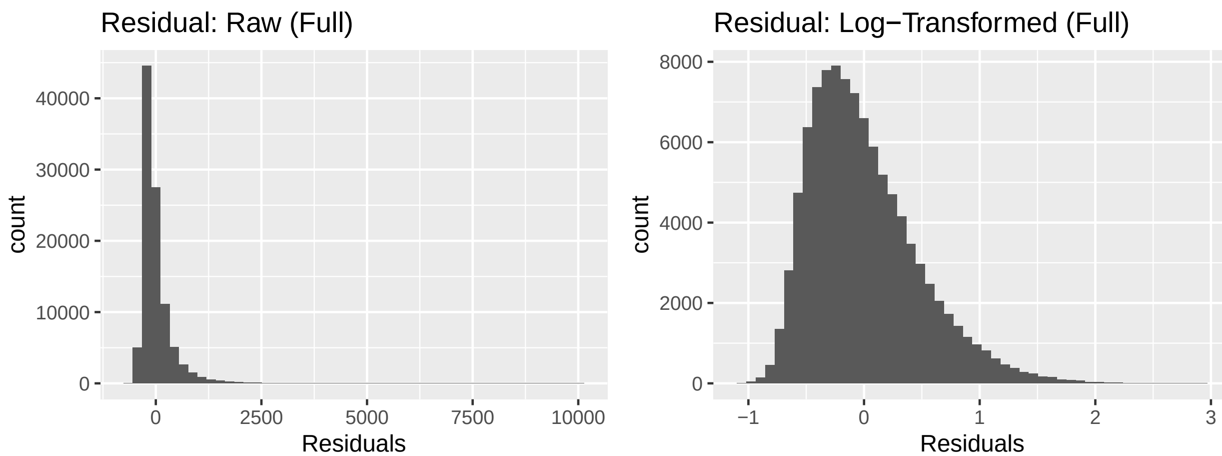

Figure 1: Left: Right-skewed distribution of raw premiums. Right: Approximate normal distribution after log-transformation.

2.2 Justification and Other Assumptions

To ensure our inference is valid, we must verify the assumptions for both OLS regression and the ANOVA F-tests used for model comparison. These assumptions largely overlap and will be assessed in the Results section using residual diagnostics:

OLS: 1. Linearity: The relationship between predictors and the log-response must be linear. We will assess this by examining the Residuals vs. Fitted plot for random scatter.

2. Independence: Observations must be independent. While we assume this holds for the analysis, we note that the voluntary nature of the survey data may introduce selection bias.

3. Equal Variance (Homoscedasticity): The variance of residuals must be constant across all fitted values. We will verify this by analysing the Residuals vs. Fitted plot for constant variance.

4. Normality: OLS requires that the error terms are normally distributed. We have earlier assessed this by analysing residual histograms.

ANOVA: 1. ANOVA used to compare models only requires that the two models compared are nested. DRF is nested in full and MLC is also nested in full.

3. Results

3.1 Diagnostics and Assumptions Checking

To ensure the validity of our inference, we examined the residual plots for the full Model (Figure 2, Left), DRF model (Figure 2, Middle), and MLC Model (Figure 2, Right) as well as data collection procedures.

- Linearity: Linearity is satisfied for the Full and DRF model. However, in the MLC model, there is a noticeable gap in the residual plots at around the 6.3 fitting value, which technically breaks the assumption of linearity.
- Constant Variance: The residuals vs. fitted plot shows that points are roughly evenly spaced out in the y direction regardless of the predicted value for DRF and full model (however there is a slightly large concentration of points above the x axis than below). This indicates somewhat equal variance for DRC and the full model. For MLC a small "valley" may have resulted in less even variance for each predicted value.
- Independence: One key source of this dataset [4] is the National Health Interview survey; this survey is conducted on a voluntary basis [7]. Consequently, there is a self-selection bias that may affect the ability to fully assume independence for this study. Further, confounding variables may exist. For instance, variables such as location may be affecting bmi and smoking. However, this study will assume that the effects of these factors are minimal.
- Normality of Residuals: As shown earlier in the residual histograms, the normality of the residuals for each of the three models is roughly fulfilled.

3.2 ANOVA Model Comparison

To test whether the addition of specific variable categories significantly improves the model, we utilized ANOVA F-tests.

Table 2: ANOVA Comparison: Demographics vs. Full Model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
99980	22900.88	NA	NA	NA	NA
99974	22087.95	6	812.93	613.24	0

Table 3: ANOVA Comparison: Lifestyle vs. Full Model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
99993	22683.72	NA	NA	NA	NA
99974	22087.95	19	595.76	141.92	0

Testing the Addition of Demographic Factors

H(null): $\beta_{Age} = \beta_{Sex} = \beta_{Region} = \beta_{Income} = \beta_{MaritalStatus} = \beta_{Urban/Rural} = \beta_{Education} = 0$ (Demographics add no explanatory power)

H(alt): At least one of $\beta_{Age}, \beta_{Sex}, \beta_{Region}, \beta_{Income}, \beta_{MaritalStatus}, \beta_{Urban/Rural}, \beta_{Education} \neq 0$

We compared the Lifestyle-only model against the Full model to see if adding Demographics significantly reduced the squared residuals. The F-test yielded an F-statistic of 613.24 and a p-value <0.001, which is less than = 0.05. We reject the null hypothesis and conclude that at least one of the coefficients of DRF has a statistically significant difference from zero. This means adding the DRF variables results in a higher R^2. This confirms that demographic factors (Age, Income, etc.) do provide statistically significant explanatory power.

Testing the Addition of Modifiable Lifestyle Factors

H(null): $\beta_{Smoker} = \beta_{Alcohol} = \beta_{BMI} = 0$ (Lifestyle Choices add no explanatory power)

H(alt): At least one of $\beta_{Smoker}, \beta_{Alcohol}, \beta_{BMI} \neq 0$

We compared the Demographic-only model against the Full model to see if adding Lifestyle factors improved the fit. The F-test yielded an F-statistic of 141.92 and a p-value <0.001. As the p value is less than = 0.05, we reject the null hypothesis and conclude that at least one of the coefficients of MLC has a statistically significant difference from zero. This means adding the MLC variables selected results in a higher R^2. This confirms that modifiable choices (Smoking, BMI) also significantly explain variation in premiums.

The ANOVA tests conclude that the addition of MLC variables to a DRF model and the addition of DRF variables to an MLC model helps explain variations of the log(annual premium).

3.3 Comparison of R^2 Values

While ANOVA tests for statistical significance, comparing R^2 values allows us to assess the practical explanatory power (effect size).

Table 4: R^2 Values of Demographic, Lifestyle, and Full Model

Category	R2_Value
R^2 Demographic	0.0249
R^2 Lifestyle	0.0342
R^2 Full Model	0.0595

Table 5: Marginal Contribution to R^2

Category	R2_Value
Change in R^2 (Full - Demographic)	0.0346
Change in R^2 (Full - Lifestyle)	0.0254

The R^2 values are generally low across all models (0.0249 to 0.0595), indicating that the variables available in this dataset explain only a small fraction of the total variance in premiums.

However, we can also compare the relative impact of the two categories. The removal of MLC from the full model decreased the R^2 by roughly 0.0346. Conversely, the removal of DRF from the full model increased the R^2 by only 0.0254. This suggests that modifiable lifestyle factors have a stronger marginal effect on predicting insurance costs than fixed demographic factors.

3.4 Regression Coefficients

Table 6 presents the coefficients for bmi, smoker status, alcohol usage, age, and education level for the Full Model. These coefficients were specifically included as the p values associated were surprising or significant.

Table 6: Coefficients for MLC and DRF variables (Model 3)

Predictor	Estimate	Std. Error	P-Value
AGE	0.00480	0.0000930	<0.001
Education: HS	-0.00974	0.0072152	0.177
Education: Some College	-0.00955	0.0072083	0.185
Education: Bachelors	-0.01356	0.0071448	0.058
Education: Masters	-0.00512	0.0076781	0.505
Education: Doctorate	-0.02424	0.0108568	0.026
Region: East	-0.00090	0.0054174	0.868
Region: North	0.00307	0.0053219	0.563
Region: South	0.00963	0.0051163	0.060
Region: West	0.00541	0.0055362	0.328
BMI	0.00562	0.0002976	<0.001
Smoker: Former	0.07895	0.0039164	<0.001
Smoker: Current	0.26276	0.0046255	<0.001
Alcohol: Occasional	-0.00063	0.0034996	0.857
Alcohol: Weekly	-0.00388	0.0042998	0.367
Alcohol: Daily	0.01258	0.0071758	0.080

Key Findings:

1. Smoking: When controlling for other variables, a current smoker and a former smoker has a p value that is statistically significant ($p < \alpha = 0.05$). Notably a "Current Smoker" has a large positive coefficient (0.263). Holding other factors constant, current smokers pay approximately 0.26 more in log(annual premiums) than never-smokers at the baseline age.
2. BMI: The coefficient for BMI is statistically significant ($p < \alpha = 0.05$) from 0. The model predicts that, holding all other variables constant, for each increase in one BMI point, the log(annual premium) increases by 0.00562.

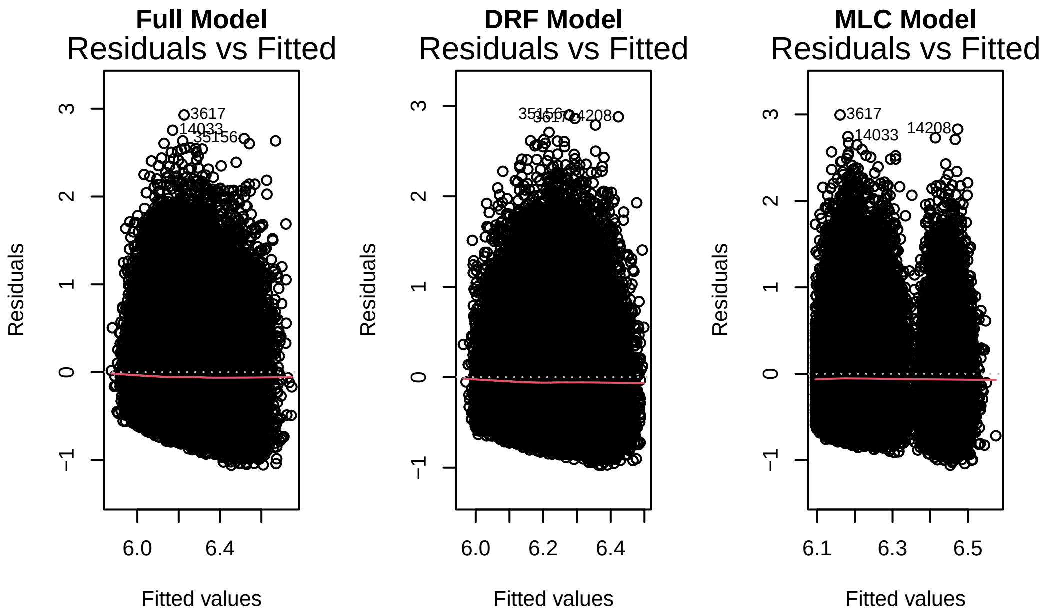

Figure 2: Diagnostic plots for the Full Model. Left: Residuals vs. Fitted values showing constant variance. Right: Normal Q-Q plot showing approximate normality.

3. Alcohol: When controlling for other variables, the p-values for alcohol frequency (occasional, weekly) are $> \alpha = 0.05$. This suggests that, unlike smoking and BMI, self-reported alcohol frequency is not a statistically significant predictor of log(annual premiums) in this dataset.
4. Age: When controlling for other variables, p values were ($p < 0.05$), indicating that there is a statistically significant non-zero linear association between log(annual premiums) and age.The model predicts that, holding all other variables constant, for each increase in one year of age point, the log(annual premium) increases by 0.00480.
5. Education: When controlling for other variables, p values were ($p < 0.05$) for only those holding doctorates. This indicates that only those holding doctorates had a statistically different log(annual premium) compared to those with no HS education. The model predicts those with doctorates to have -0.02424 lower log(annual premium) values compared to those with no HS education.
6. Region: When controlling for other variables, p values were ($p > 0.05$) for all regions. This suggests that being in the north, east, south, and west does not lead to a significantly significant change in log(annual premium) compared to those who lived in the central US.

4. Discussion

4.1 Results Interpretation

Initial analysis using multiple hierarchical multiple linear regression showed that the addition of modifiable lifestyle choice variables (MLC) and demographic factors (DRF) significantly affect the log(annual insurance). Further, comparison of R^2 of MLC-only vs. full model and DRF-only vs. full model indicated that MLC variables selected may have a more significant impact on improving fit (by minimizing residuals squared) of the regression line.

Analysis of the regression coefficient revealed several notable pieces of information. When holding other variables constant, smoking, bmi, age, and whether the individual has earned a doctorate was associated with higher log(annual premiums). The slope value associated with smoking was high, at 0.22. As smoking and bmi are MLC variables, individuals seem to have some control in reducing their healthcare cost. Individuals interested in reducing healthcare costs can do so by reducig their BMI score and quiting smoking. Most of these results in MLC align with our expectations. The Affordable Care Act (ACA) allows smoking status to factor into healthcare costs. Further, high BMI and age can increase likelihood of other health conditions, that lead individuals to select costlier health insurance. Lower age and holding a doctorate are DRF variables associated with lower healthcare costs. Holding a doctorate may be associated with higher income compared to those without HS education, thus, perhaps leading to healthier lifestyles. Age's association with higher helthcare may indicate the need for a systemic change to make healthcare more affordable to the elderly.

There is insufficient evidence to show that alcohol consumption frequency and region was associated with log(annual premiums). While this result is expected for alcohol consumption as it is not factored into insurance costs, it is surprising for region as region factors into costs [3].

4.2 Methodological Limitations

Several issues limit the strength of these conclusions. Due to a limitation of the dataset, we are unable to consider, and hold constant, the type of insurance plan selected. This variable is allowed to insure determine cost under the ACA. Further, other variables that are associated with Modifiable Lifestyle Choices and Demographic factors, such as history of cancer and ethnicity, are not included. These missing variables may account for the low R^2 values of our fitted models. Further, generalizing to all MLC and DRF in general is not reasonable. Second, self-reported data is utilized by the NHIS, which is a large source for datapoints in our datasets [7]. Consequently individuals may under report factors such as BMI, smoking and alcohol consumption due to stigmatization. As our data does not perfectly align with the assumptions for linear models, estimates may also not be fully accurate.

4.3 Recommendations

To improve this study, more variables representing MLC and DRF should be included. It may also be helpful to test the interaction effects between MLC and DRF variables, as variables such as bmi and income may be associated. Second, a new dataset that does not rely on self reported data and insurance plan types would work better. Datasets that subdivide variables such as alcohol consumption and region into more categories may be helpful. There may be variations within existing categories: an individual in South Carolina versus North Carolina may have very different insurance premium costs.

5. References

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